

3'end RNA-seq using template switching protocol

Status: [?] | partially migrated / unconfirmed

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About

Protocol for 3' end RNA-seq using a template switching reverse transcription workflow.

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1. RNA extraction

Input / starting material: RNA lysate

- Maxwell RNA extraction
 - Alternative: Add 3 volumes of LS-Trizol to lysate
 - Extract RNA using the Zymo MicroSpin Columns
 - Elute with 6 μ L of H₂O
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2. RNA fragmentation and template switching RT

- Prepare fragmentation buffer.
- Fragmentation buffer: 80 mM Tris-HCl pH 7.5, 40 mM MgCl₂.

Note: have more recently been using milder fragmentation buffer conditions with input concentration of 10 mM Tris-HCl pH 7.5, 5 mM MgCl₂ (final concentration 1.25 mM Tris-HCl and 0.625 mM MgCl₂).

- Use 500 ng of RNA (whenever possible).

- Bring to 3.5 μL using water.
 - Add 0.5 μL of fragmentation buffer (10 mM Tris-HCl pH 7.5, 5 mM MgCl final concentration).
 - Fragment RNA by incubating 12 min at 95 $^{\circ}\text{C}$.
 - Add:
 - 0.5 μL 10 mM dNTPs
 - 0.5 μL 5 μM primer RT1
 - Mix and incubate at 65 $^{\circ}\text{C}$ for 3 min then cool down to 42 $^{\circ}\text{C}$ and hold.
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3. Template switching RT incubation and cleanup

- Make a mastermix and add 5.5 μL of the following:
 - 2 μL of SSIV buffer
 - 2 μL 5 M betaine
 - 0.5 μL 0.1 M DTT
 - 0.25 μL 40 μM TSO
 - 0.25 μL 200 mM MgCl
 - 0.25 μL SuperRNasin
 - 0.25 μL SSIV
 - Place the mastermix into a PCR strip and into the cycler so that it pre-warms to 42 $^{\circ}\text{C}$ before you start adding it to the samples.
 - Add mastermix to samples while keeping them at 42 $^{\circ}\text{C}$ in the cycler (i.e., don't take the samples out of the cycles while adding the mix).
 - Incubate at 42 $^{\circ}\text{C}$ for 10 min.
 - Incubate at 50 $^{\circ}\text{C}$ for 30 min.
 - Purify using Magbind beads (1.8 μL per 1 μL of PCR reaction). Elute in 8 μL H₂O.
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4. Optional PCR test

- Use 1 μL RT product.
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5. Indexing PCR

- Prepare the following 20 μL indexing PCR reaction for each sample:
 - 10 μL of 2 $\times$ Q5
 - 2 μL RT product
 - 1 μL dual indexed primers

- * Use a different index for each sample.
 - * Make sure each sample will have a different index if submitting multiple libraries at the same time.
 - 7 μL of ddH₂O
 - Incubate for the PCR reaction using the Q5 program:
 - Extension time – 45 seconds
 - T_m – 65 °C
 - Typically try 10 and 14 cycles
 - Move samples to post-PCR room.
 - Add 2 μL of purple 6X loading buffer to 5 μL of PCR reaction. Keep the rest of the 20 μL .
 - Load on a 6% TBE gel together with 2 μL of ready-to-use low molecular weight ladder (NEB).
 - Run at 180 V for 25 min.
 - Stain with Sybr Green (1:10000 dilution) in 1X TBE.
 - Visualise.
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6. PCR beads purification

- Mix together approx. 10 μL of each PCR reaction.
 - Add a suitable amount of Magbind beads (e.g. 0.9 μL per 1 μL of PCR reaction) and mix.
 - Incubate at room temperature for 5 minutes.
 - Remove supernatant and wash with 500 μL of 70% Ethanol for 30 sec. Repeat this step.
 - Air dry the beads.
 - Resuspend in 30 μL of H₂O.
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7. Buffers

- Fragmentation buffer: 80 mM Tris-HCl pH 7.5, 40 mM MgCl₂.
 - Milder fragmentation buffer conditions: input concentration of 10 mM Tris-HCl pH 7.5, 5 mM MgCl₂ (final concentration 1.25 mM Tris-HCl and 0.625 mM MgCl₂).
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